

First-year Analysis Examination
Part One
May 2025

Answer FOUR questions in detail.
State carefully any results used without proof.

1. Let A be a subset of the metric space M . Decide (with proof or counterexample) whether the following statements are equivalent: (i) A is open; (ii) for each $X \subseteq M$, if $A \cap X = \emptyset$ then $A \cap \overline{X} = \emptyset$.
2. The function $f : X \rightarrow Y$ between metric spaces is distance-preserving. Must $f(X)$ be closed in Y if X is (i) compact, (ii) complete? If the answer is Yes, give a proof; if the answer is No, give a counterexample.
3. Let $f : X \rightarrow Y$ be a continuous bijection, $g : Y \rightarrow X$ its inverse. Must g be continuous (i) if X is compact, (ii) if Y is compact? Proof or counterexample, as appropriate.
4. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and satisfies

$$\liminf f(n) < 0 < \limsup f(n)$$

where n runs through the natural numbers. Prove that each of $f^{-1}(-\infty, 0)$ and $f^{-1}(0, \infty)$ is infinite.

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that

$$|f(x) - f(y)| \leq |x - y|^a$$

for some $a > 1$ and all $x, y \in \mathbb{R}$. Deduce as much as possible about f .
